Avinash Kolhe

Full Stack Software Engineer | AI / ML Engineer

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PROFESSIONAL SUMMARY

Full-Stack software and AI/ML Engineer with hands-on experience in building intelligent, scalable applications using Agentic AI, computer vision, and OCR. Skilled in MERN Stack, Python, FastAPI, and REST API development, with a strong ability to turn complex ideas into practical solutions. First Runner-Up at the ICAIM GenAI competition, driven by innovation, problem-solving, and continuous learning.

TECHNICAL SKILL

Programming Language: SQL, JavaScript, Python, Java (DSA).
Web Frameworks & Technologies: MongoDB, Express.js, React, Node.js (MERN Stack).
AI / ML Engineering: Retrieval-Augmented Generation (RAG), LLM Orchestration, Semantic Search, Vector Embeddings, Prompt Engineering, Computer Vision, Optical Characters Recognition (OCR).
Software Engineering: Data Structures & Algorithms, Object-Oriented Programming (OOP), RESTful API Design, Authentication & Authorization, Role-Based Access Control (RBAC), Multi-Tenant Architecture, Event-Driven Architecture, Asynchronous Programming.
Backend & Database Concepts: Database Design, Query Optimization, Data Indexing, Data Modelling, Database Indexing, API Integration, Real-Time Communication, Secure API Development, Integration Testing.
Development Practices: Version Control, Debugging(Git), Unit Testing, API Testing, Agile Methodology, Software Development Life Cycle (SDLC).

PROJECTS

Agentic AI – AI Document Intelligence & Operations System (github.com/avinashkolhe066/my_rag_project)

- Developed a full-stack AI-powered document intelligence platform using React, Node.js, Express, and Python FastAPI, supporting document-based Q&A across PDF, CSV, JSON, and TXT formats.
- Integrated Agentic AI capabilities with Ollama (Qwen2.5:7B) and Gemini API for intelligent document processing, citation-based responses, and automated quiz generation.
- Implemented JWT authentication, role-based access control, Cloudinary storage, Stripe payments, automated certificate generation, and dynamic PDF report generation using modular REST APIs.

EduVerify – AI-Powered handwriting Recognition System (github.com/avinashkolhe066/Handwriting-Recognition)

- Developed a full-stack MERN application with Python FastAPI microservices for digital assignment recognition across three roles: Student, Admin, and Super Admin, later extended to React Native.
- Built a three-stage AI verification pipeline using Microsoft TrOCR, Meta DINOv2, and ChromaDB for handwriting recognition, authenticity verification, and plagiarism detection.
- Implemented JWT-based authentication, real-time Socket.IO notifications, and an Ollama-powered AI admin assistant to automate grading and streamline feedback workflows.

EDUCATION

Master Of Computer Applications (MCA)	University Of Mumbai	7.92 SGPA	2027
Thakur institute of management Studies, Career Development & Research.			
Bachelor Of Computer Application (BCA)	YCMOU	7.38 CGPA	2025
Yashwantrao Chavan Maharashtra Open University.			

CERTIFICATION PROGRAM

- Generative AI Certification – IBM (February 2026)
- NPTL – The Joy of Computing Using Python
- NPTL – Cloude Computing

ACHIEVEMENTS & ACTIVITIES

- GenAI Competition — First Runner-Up, ICAIM event (competitive AI/ML build-and-pitch challenge).
- Participant, IndiaNext 2026 Hackathon — collaborated in a team to build a software solution under time constraints.
- Active member, College Coding Club — contributes to collaborative software development and technical projects

LANGUAGES & INTERSTS

Languages: English, Marathi, Hindi **Interests**: Vibe Coding, PC Gaming, Traveling, Tracking